

Java Program: NullPointerException Handling

1. Aim

To write a Java program that demonstrates a NullPointerException and handles it using a try-catch block.

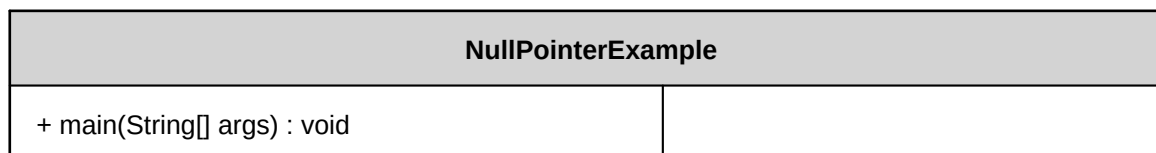
2. Description

A NullPointerException occurs when a program tries to use a null object reference as if it referred to a valid object. In this program, the String variable **name** is assigned the value null. Printing the variable is valid and displays the word null. However, calling the **length()** method on the null reference causes a NullPointerException. The catch block catches the exception and displays suitable messages instead of allowing the program to terminate unexpectedly.

3. Algorithm

1. Start the program.
2. Enter the main method.
3. Declare a String variable named name and assign null to it.
4. Display the value of name.
5. Try to find the length of name using the length() method.
6. Since name is null, a NullPointerException is generated.
7. Catch the exception using a catch block.
8. Display messages explaining that a null object cannot be used for the operation.
9. Stop the program.

4. UML Diagram



5. Source Code with Comments

```
public class NullPointerException {
    public static void main(String[] args) {

        try {
            // Declare a String reference and assign null to it
            String name = null;

            // Printing a null reference is allowed
            System.out.println("Name: " + name);

            // Calling a method on a null reference causes
            // a NullPointerException
            System.out.println("Length: " + name.length());

        } catch (NullPointerException e) {
            // Handle the NullPointerException
            System.out.println("Caught NullPointerException!");
            System.out.println("Cannot perform an operation on a null object.");
        }

    }
}
```

6. Compilation and Execution Commands

```
javac NullPointerException.java
java NullPointerException
```

The first command compiles the Java source file. The second command executes the compiled program.

7. Output Screenshot

Program Output

```
Name: null
```

```
Caught NullPointerException!
```

```
Cannot perform an operation on a null object.
```

8. Result

The Java program was successfully executed. It demonstrated a `NullPointerException` when a method was called on a null `String` reference, and the exception was successfully handled using a try-catch block.